

UNIVERSITY OF ASIA PACIFIC
DEPARTMENT OF CSE

Lab Report - 7

COURSE CODE: CSE 402.

COURSE TITLE: OPERATING SYSTEM LAB.

SUBMITTED BY:

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SUBMITTED TO

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1. Problem:

Given the following disk request queue and initial disk head position:

- Request Queue: [3, 60, 10, 25, 30, 4]
- Initial Head Position: 41

Implement the following Disk Scheduling Algorithms:

- Shortest Seek Time First (SSTF)
- First Come First Serve (FCFS)

For each algorithm, calculate the total disk head movement and represent the request servicing sequence. Finally, compare the performance of SSTF and FCFS based on total head movement and determine which algorithm is more efficient for the given request queue.

2. Source Code:

```
req = [3, 60, 10, 25, 30, 4]
head = 41
n = len(req)

done = [0] * n
current = head
total_movement = 0
seq = [current]

for i in range(n):
    x = -1

    min_distance = 9999

    for j in range(n):
        if done[j] == 0:
```

```
            dist = abs(req[j] - current)

            if dist < min_distance:
                min_distance  = dist
                x = j

    movement = min_distance


    print(f"{current} -> {req[x]} = {movement}")

    total_movement += movement
    current = req[x]
    seq.append(current)
    done[x] = 1

print("Total Movement: ", total_movement)
print("Sequence" , seq)
print()

# FCFS disk Schedulig

req = [3, 60, 10, 25, 30, 4]
head = 41

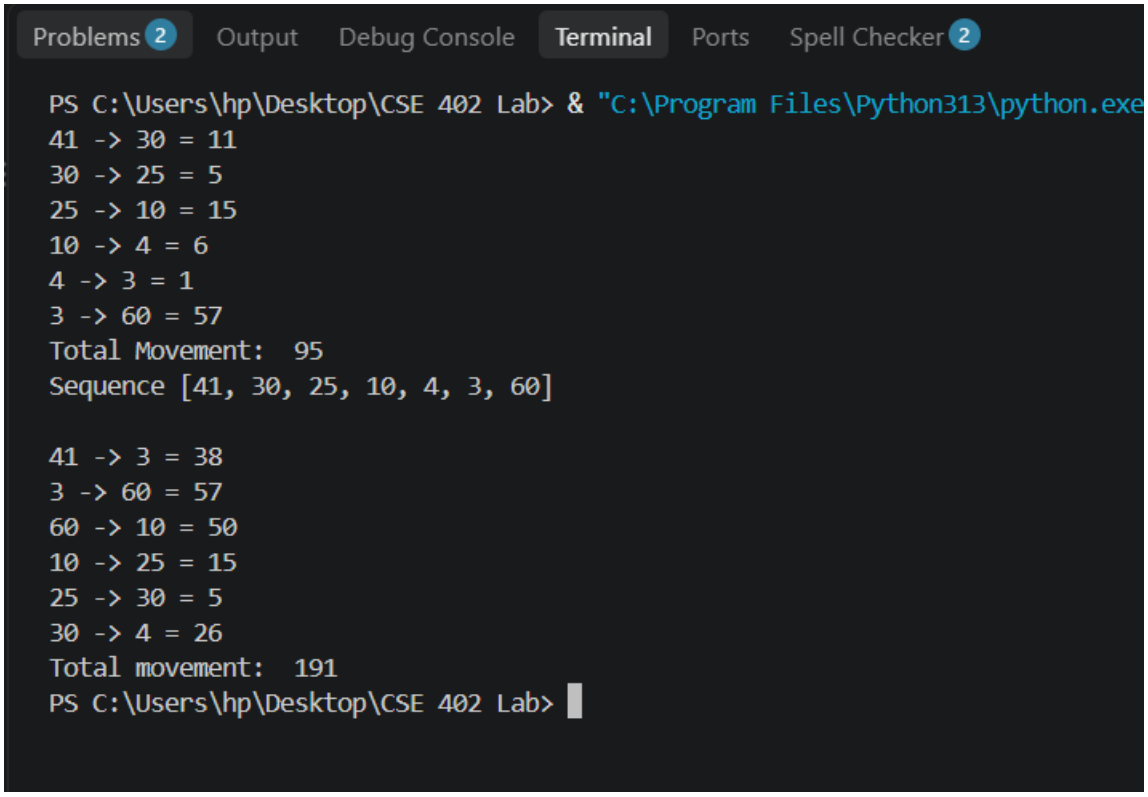
total_m = 0
current = head
seq = [current]


for req in req:
    movement = abs(req - current)
    print(f"{current} -> {req} = {movement}")

    total_m += movement
    current =  req
```

```
print("Total movement: ", total_m)
```

3. Output:



4. Discussion:

This experiment compares SSTF and FCFS disk scheduling algorithms using the same request queue and initial head position. The main objective is to evaluate their efficiency based on total disk head movement.

The results show that SSTF requires 95 tracks of total head movement, whereas FCFS requires 191 tracks. FCFS processes requests in their original order, which can cause large movements between consecutive requests. In contrast,

SSTF selects the closest pending request at each step, reducing the total distance travelled by the disk head.

SSTF reduces the total head movement by 96 tracks, approximately 50.2618% compared with FCFS. Therefore, SSTF performs better for this particular request queue.

5. Conclusion:

The experiment successfully implemented and compared the SSTF and FCFS disk scheduling algorithms. SSTF achieved a total head movement of 95 tracks, whereas FCFS required 191 tracks to service the same requests. Therefore, SSTF performs better for this particular input by reducing disk head movement by 96 tracks, or approximately 50.2618%. Although FCFS follows a simple and predictable request order, SSTF provides better performance when minimizing total head movement is the main objective. However, SSTF may cause starvation for requests that remain far from the current head position.

6. GitHub:

https://github.com/Rizon40/CSE-402-Operating-System-Lab